PAGE :
DATE : / /

# UNIT - 4

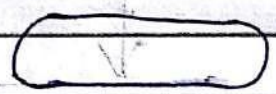
PAGE:
DATE: / /

**Algorithm:-** To write a logical step by step method to solve the problem is a ~~pr~~ called algorithm. In other words algorithm is a processor for solving problems. An Algorithm include Calculations, Reasoning and data processing.

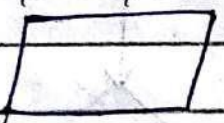
**Flowchart:-** A flow chart is the graphical or pictorial representation of an Algorithm. with the help of different Symbols. with algorithms we can usualy understand a program the main purpose of a flowchart is to analy's different process.

**Flowchart:-**

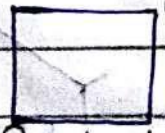
* Terminal Box (Start and End).



* Input / Output



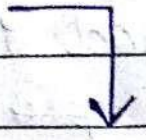
* Process / Instructions



* Decision

PAGE:
DATE: / /

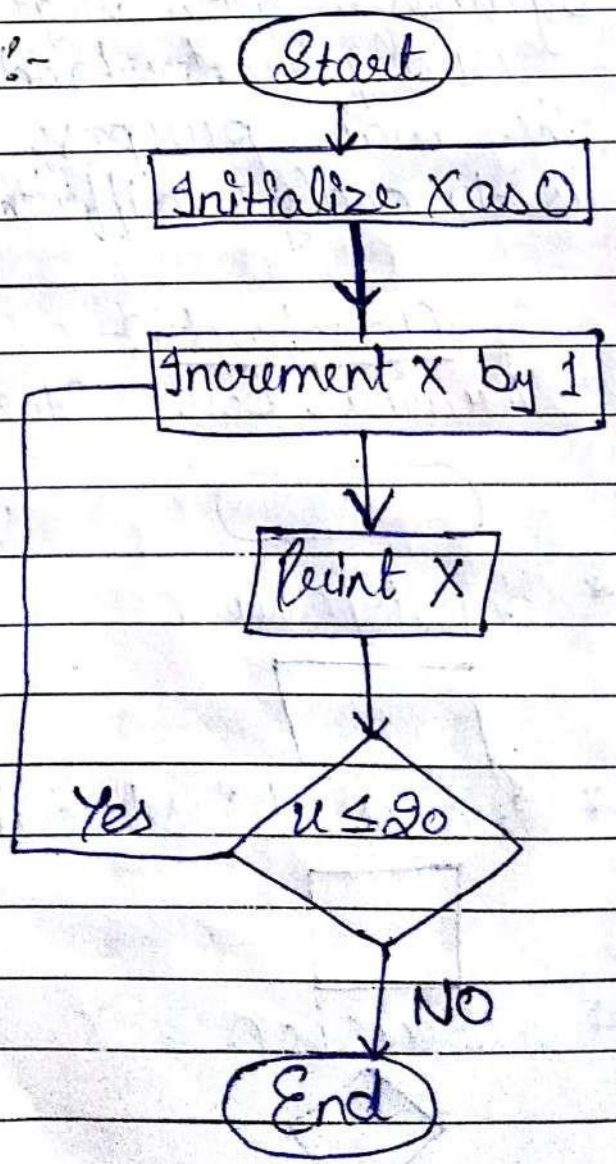
* Connector / Arrow



Example :-
Write a Algorithm to print 1 to 20

Algo :-
Step 1 :- Initialize X as 0,
Step 2 :- Increment X by 1,
Step 3 :- Print X,
Step 4 :- If x is less than or equal to 20 then go back to step 2.

Flowchart :-



PAGE :
DATE : / /

Program 12 :- W.A.P to find the sum of first n ~~no~~ numbers :-

```
#Include <stdio.h>
#Include <conio.h>
    Void main ( )
    {
  clrscr ( );
 int n, i, sum;
 Sum = 0;
 print ("Enter the value of n");
 Scanf ("%d", &n);
 for (i=1; i<=n; i++).
  {
 Sum = Sum + i;
  }
 printf ("Sum of first n numbers : %d",
                    sum);
 getch ( );
  }
```

Output :-

enter the value of n 5

Sum of first n numbers is 15

PAGE:
DATE: / /

Program 13:- W.A.P. to print the following pattern

```
* * * *
* * * *
* * * *
* * * *
```

```
# Include <stdio.h>
# Include <conio.h>
  void main ( )
  {
  clrscr ( );
int i, j;
for (i=1; i<=4; i++)
  {
  for (j=1; j<=4; j++)
    {
printf ("*");
    }
printf ("\n");
  }

getch ( );
  }.
```

Output:-

```
* * * *
* * * *
* * * *
* * * *
```

PAGE:
DATE: / /

Program 14:- W.A.P. to print the following pattern:-

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

```
#Include <stdio.h>.
#Include <conio.h>.
void main ( )
{
clrscr( );
int i, j;
for(i=1; i<=6; i++).
{
for(j=1; j<=i; j++)
{
printf ("*");
}
printf ("\n");
}.
getch ( );
}
```

Output:-

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

PAGE:
DATE: / /

* **Divide and Conquer Algorithms :-**
In computer science, divide and conquer algorithm work by breaking down a problem into two or more sub-problems of the same time until these become simple to be solve directly. The solution to subproblem are then combined to give a solution to the original problem.
This technique is the basis of efficient algorithms for all kinds of problems such as sorting.
ex:- Quick Sort, Merge Sort

* **Trail and error algorithm :-**
Some complex problem can be solve by a technique that is called trail and error.
Trail and error is typically good for problems where you have multiple chance to get the correct solution.
Trail and error is not a method of finding the best solution is it a method of finding all solutions.
One of the most powerful advantage of these techniques is that it does not require you have a

PAGE :
DATE : / /

lots of knowledge. Trail and error is used to discover new things and it also play and important role in the scientific method as well.

* Steps in problem solving :-

There are seven main steps to follow when trying to solve a problem. These steps are as follow -

1) Define and Identify the problem.
2) Analyze the problem.
3) Identify possible Solutions
4) Selecting the best Solutions
5) Evaluating the Solution
6) Develop an action plan.
7) Implement the Solution.

1- Define and Identify the problem - The first step is critical. It is essential for each group member to clearly understand the problem, so that all energy will be focused in the same direction. A good way to define the problem is to write down a concise statement which summerize the problem.

2- Analyze the problem - In this stage of problem solving, questions should be

PAGE:
DATE: / /

asked and Information gathered. Think about how the issue affects others. It is essential to spend more time researching the problem.

3- Identifying Possible Solutions - Identifying possible solutions to the problem is sometimes referred to as finding "optional solutions", because the goal is to complete a list of all conceivable alternatives to the problem. Using the variety of creative techniques, group participants create an extensive list of problem solution.

4- Selecting the best Solutions - The starting point in any conscious attempt at rational decision making must be the recognition that a problem exists. While effective meetings are essential to getting work done, most meetings leave us still looking for a decision.

5- Evaluating Solutions - There are several ways to evaluate the chosen solutions. It is often helpful to make a T-Chart and ask members of the group to

PAGE:
DATE: / /

name the prose and cons of each solutions. This method will visually illustrate the strengths and weakness of each solution.

6- Develop an action plan - An action plan is a chart that lists the tasks that need to be done and identifies who will be responsible for each, when and what action is necessary, where to start, and how.

7- Implement the solution - Sometimes the groups who choose the solution are not the ones who will implement it. If this is the case, members who select the solution should clearly explain why they selected it to the ones who will implement it.